

Shell scripting tutorial sheet

1. Write a program to reverse an integer
2. Write a program to check whether an integer is Armstrong number or not.
3. Write a program to check given number is prime or not.
4. Write a program to print the Fibonacci series using iteration.
5. Write a program to print the Fibonacci series using recursion.
6. Write a program to check whether a number is palindrome or not using iteration.
7. Write a program to check whether a number is palindrome or not using recursion.
8. Write a program to find greatest among three integers.
9. Write a program to check if a number is binary.
10. Write a program to find sum of digits of a number using recursion.
11. Write a program to swap two numbers without using third variable.
12. Write a program to swap two numbers using third variable.
13. Write a program to find prime factors of a given integer.
14. Write a program to add two integer without using arithmetic + operator.
15. Write a program to check given number is perfect or not.
16. Program to find the Average of numbers with explanations.
17. Program to calculate factorial using iterative method.
18. Program to calculate factorial using recursion.
19. Program to check a given number is even or odd.

20. program to print first n Prime Number with explanation.
21. Program to print Prime Number in a given range
22. Program to find Smallest number among three.
23. program to calculate the power using the POW method.
24. Program to calculate the power without using POW function.
25. Program to calculate the square of a given number.
26. Program to calculate the cube of a given number
27. Program to calculate the square root of a given number.
28. program to calculate LCM of given two numbers.
29. Program to find GCD or HCF of two numbers.
30. Program to find GCD of two numbers using recursion.
31. Program to Convert Decimal Number into Binary.
32. Program to convert Decimal number to Octal number.
33. Program to check the given year is a leap year or not.
34. Program to convert Celsius to Fahrenheit.
35. Program to convert Fahrenheit to Celsius.
36. program to calculate Simple Interest with explanation.

1. Write a method which will remove any given character from a String.
2. Write a program to count occurrence of a given character in a String.
3. Write a program to check if two Strings are Anagram.
4. Write a program to check a String is palindrome or not.
5. program to check given character is vowel or consonant.
6. program to check given character is digit or not.
7. program to replace the string space with a given character.
8. program to convert lowercase char to uppercase of string.

9. program to convert lowercase vowel to uppercase in string.
10. program to delete vowels in a given string.
11. program to count Occurrence Of Vowels & Consonants in a String.
12. program to print the highest frequency character in a String.
13. program to Replace First Occurrence Of Vowel With '-' in String.
14. program to count alphabets, digits and special characters.
15. program to separate characters in a given string.
16. program to remove blank space from string.
17. program to remove blank space from string.
18. program to concatenate two strings.
19. program to remove repeated character from string.
20. program to calculate sum of integers in string.
21. program to print all non repeating character in string.
22. program to copy one string to another string.
23. Program to sort characters of string.
24. Program to sort character of string in descending order.

1. Write a program for, In array 1-100 numbers are stored, one number is missing how do you find it.
2. Write a program for, In a array 1-100 multiple numbers are duplicates, how do you find it.
3. Write a program to find first duplicate number in a given array.
4. Write a program to remove duplicate elements form array in C.
5. Write a program for, Given two arrays 1,2,3,4,5 and 2,3,1,0,5 find which number is not present in the second array.
6. Write a program for, How to compare two array is equal in size or not.
7. Write a program to find largest and smallest number in array.
8. Write a program to find second highest number in an integer array.
9. Write a program to find top two maximum number in array?
10. program to print array in reverse Order.
11. program to reverse an Array in two ways.
12. Program to calculate length of an array.
13. program to insert an element at end of an Array.
14. program to insert element at a given location in Array.
15. Program to delete element at end of Array.
16. Program to delete given element from Array.
17. Program to delete element from array at given index.
18. Program to find sum of array elements.
19. Program to print all even numbers in array.
20. Program to print all odd numbers in array.
21. program to perform left rotation of array elements by two positions.
22. program to perform right rotation in array by 2 positions.

23. Program to merge two arrays.
24. Program to find highest frequency element in array.
25. Program to add two number using recursion.
26. Program to find sum of digit of number using recursion.

Q1. Write script to print given numbers sum of all digit, For eg. If no is 123 it's sum of all digit will be $1+2+3 = 6$.

Q2. How to perform real number (number with decimal point) calculation in Linux

Q3. How to calculate $5.12 + 2.5$ real number calculation at \$ prompt in Shell ?

Q4. How to perform real number calculation in shell script and store result to third variable , lets say $a=5.66$, $b=8.67$, $c=a+b$?

Q5. Write script to determine whether given file exist or not, file name is supplied as command line argument, also check for sufficient number of command line argument

Q6. Write script to determine whether given command line argument (\$1) contains "*" symbol or not, if \$1 does not contains "*" symbol add it to \$1, otherwise show message "Symbol is not required". For e.g. If we called this script Q12 then after giving ,

Q7. Write script to print contains of file from given line number to next given number of lines. For e.g. If we called this script as Q13 and run as
\$ Q13 5 5 myf , Here print contains of 'myf' file from line number 5 to next 5 line of that file

Q8. Write script to implement getopt statement, your script should understand following command line argument called this script Q14,

Q14 -c -d -m -e

Where options work as

- c clear the screen
- d show list of files in current working directory
- m start mc (midnight commander shell) , if installed
- e { editor } start this { editor } if installed

Q9. Write script called sayHello, put this script into your startup file called .bash_profile, the script should run as soon as you logon to system, and it print any one of the following message in infobox using dialog utility, if installed in your system, If dialog utility is not installed then use echo statement to print message
:-

- Good Morning
- Good Afternoon
- Good Evening , according to system time.

Q10. How to write script, that will print, Message "Hello World" , in Bold and Blink effect, and in different colors like red, brown etc using echo command.

Q11. Write script to implement background process that will continually print current time in upper right

corner of the screen , while user can do his/her normal job at \$ prompt.

Answer: See Q17 shell Script.

Write shell script to implement menus using dialog utility. Menu-items and action according to selectmenu-item is as follows

Menu-Item	Purpose	Action for Menu-Item
Date/time	To see current date time	Date and time must be shown using infobox of dialog utility
Calendar	To see current calendar	Calendar must be shown using infobox of dialog utility
Delete	To delete selected file	First ask user name of directory where all files are present, if no name of directory given assumes current directory, then show all files only of that directory, Files must be shown on screen using menus of dialog utility, let the user select the file, then ask the confirmation to user whether he/she wants to delete selected file, if answer is yes then delete the file , report errors if any while deleting file to user.

Exit To Exit this shell script Exit/Stops the menu driven
program i.e. this script

Note: Create function for all action for e.g. To show date/time
on screen create function show_datetime().

Answer: See Q18 shell Script.

Write shell script to show various system configuration like

- 1) Currently logged user and his logname
 - 2) Your current shell
 - 3) Your home directory
 - 4) Your operating system type
 - 5) Your current path setting
 - 6) Your current working directory
 - 7) Show Currently logged number of users
 - 8) About your os and version ,release number , kernel version
 - 9) Show all available shells
 - 10) Show mouse settings
 - 11) Show computer cpu information like processor type, speed etc
 - 12) Show memory information
 - 13) Show hard disk information like size of hard-disk, cache memory, model etc
- File system (Mounted)